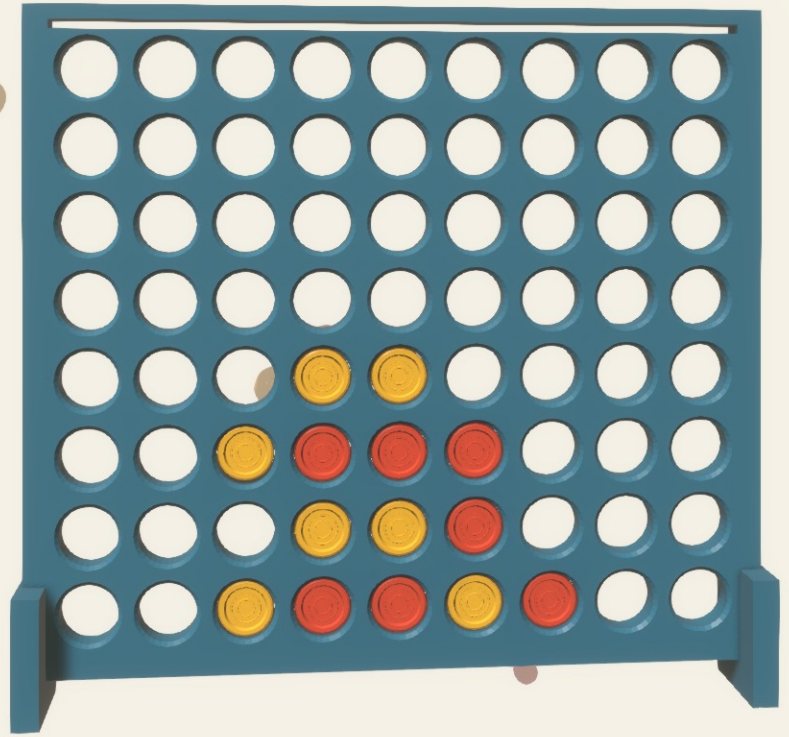
group 3 · TJ Raklovits and Andrew Roddy

Connect Four AI



the question

A board nobody has solved



9 × 8

never solved
our test board

does what they learn carry over, and why?

scale

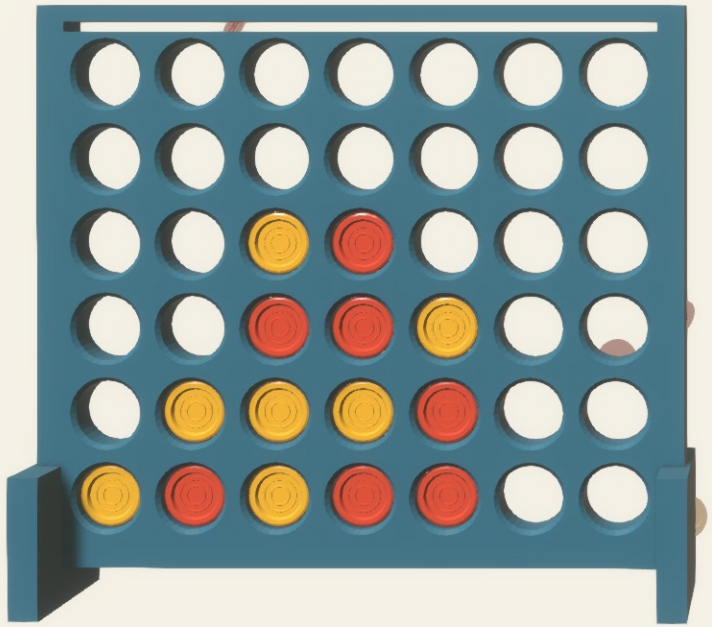
How big is the game?



An exact solver

if yellow plays in this column: win, draw or loss

W L L W W W L



UCI dataset: 67,557 positions

100%

Pascal Pons' tests: 6,000 positions

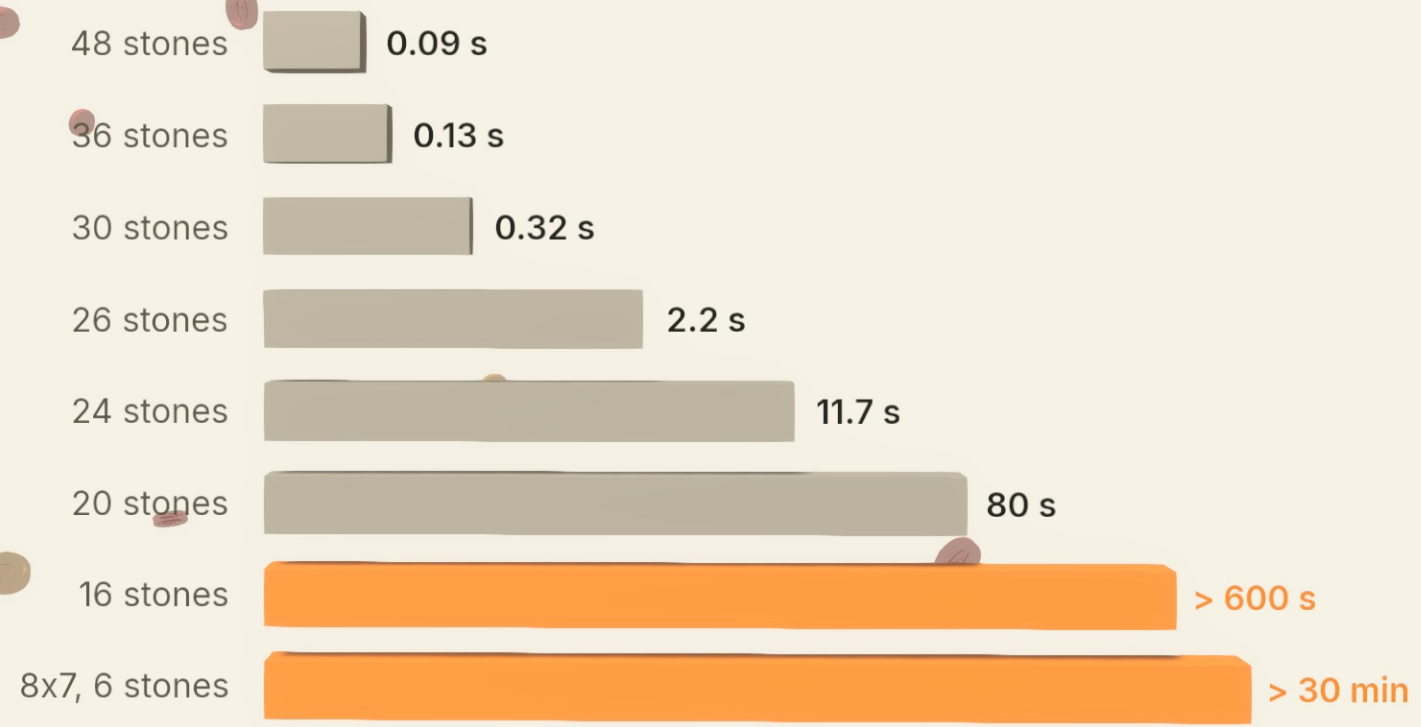
100%

0 wrong

the wall

Why not just solve 9x8?

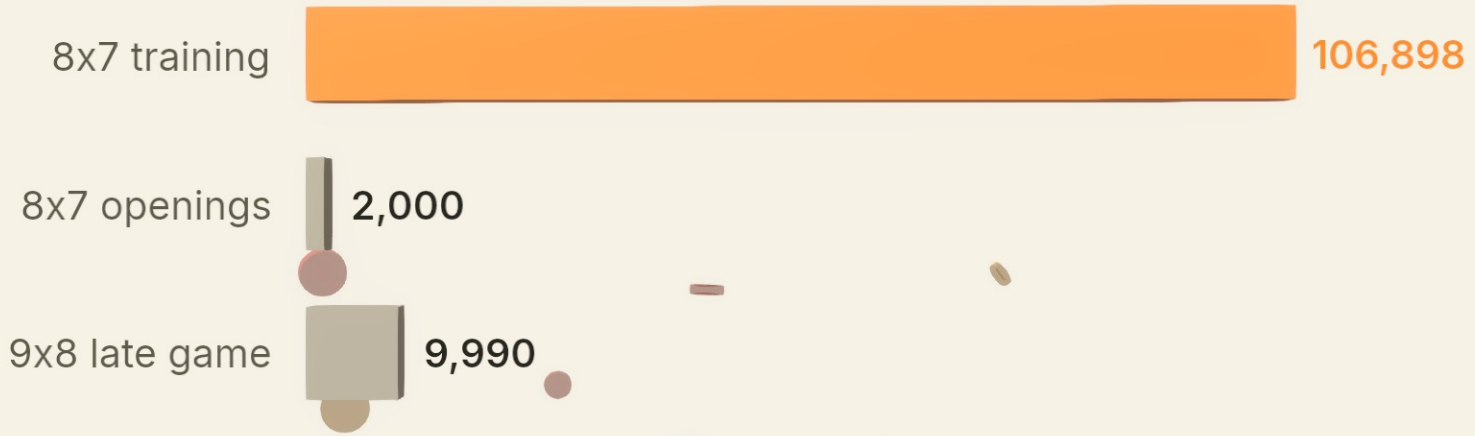
seconds to solve a position (log scale), by stones on the board



step 2

The data

positions labelled by the solver



192 cores, one position each

step 3

Eight agents

MCTS (UCT)

search · Elo 740

Alpha-beta (learned)

search + learning · Elo 722

Alpha-beta (handcrafted)

search · Elo 690

Network-guided MCTS

search + learning · Elo 647

Allis rules

knowledge · Elo 160

Exact solver

search · Elo 78

Convolutional network

learning · Elo 37

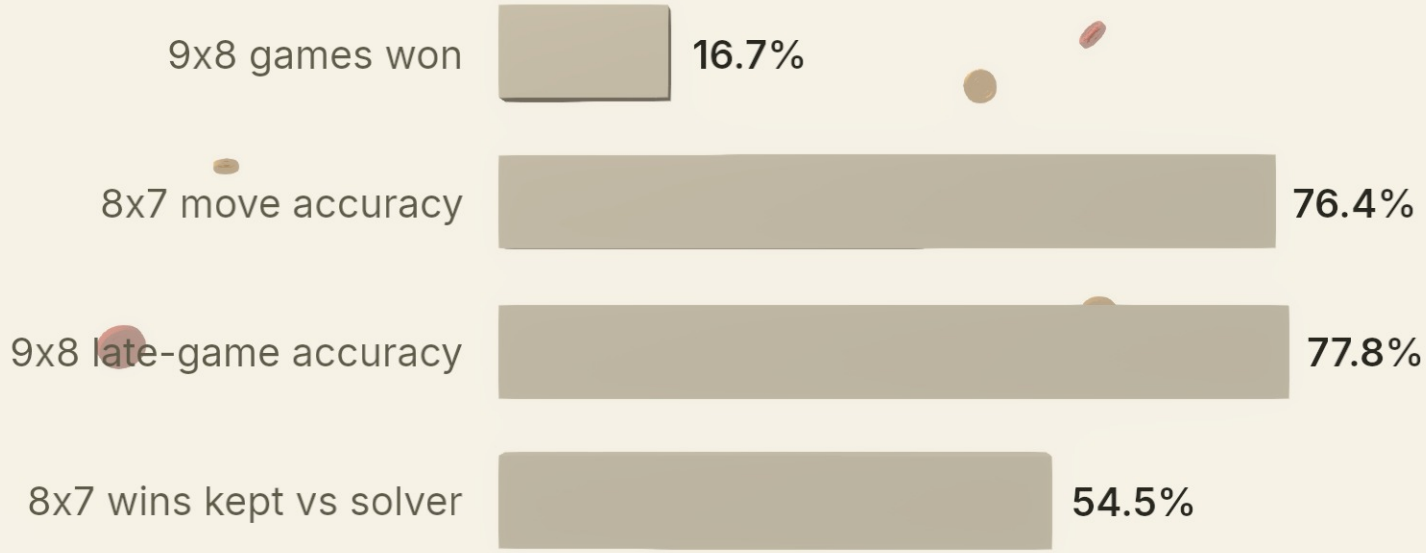
Safe random

baseline · Elo 0

agent 1 of 8 · baseline

Safe random

the same four numbers for every agent (%)



– no plan beyond the next move

Allis rules

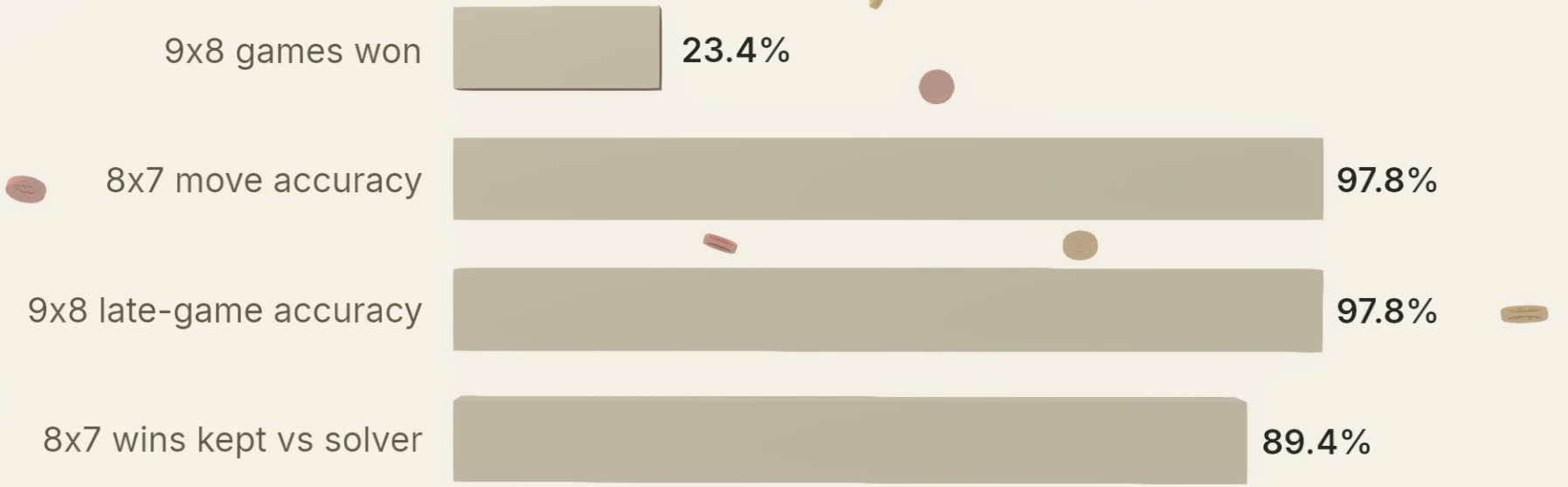
the same four numbers for every agent (%)



– no lookahead, so tactics catch it

Exact solver, 1 s a move

the same four numbers for every agent (%)



– one second is nowhere near enough

agent 4 of 8 · search

Alpha-beta, handcrafted

the same four numbers for every agent (%)



– only as good as our hand-picked weights

agent 5 of 8 · search + learning

Alpha-beta, learned

the same four numbers for every agent (%)



– a linear model can't find new ideas

Monte Carlo tree search

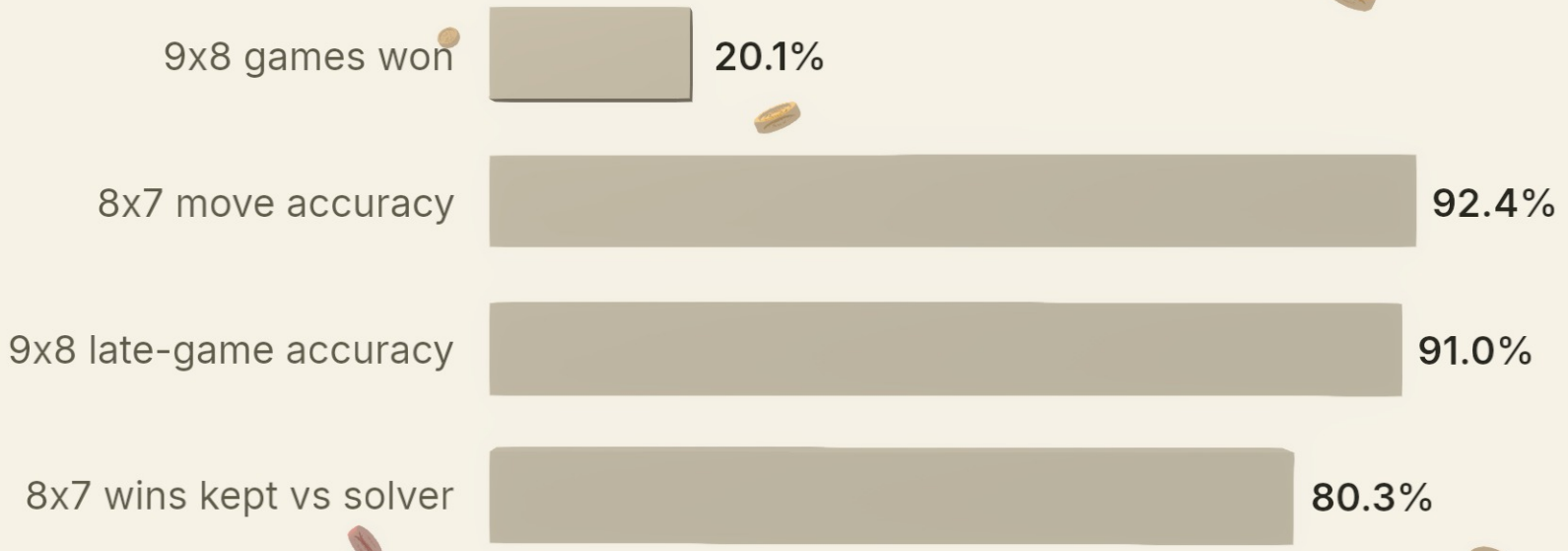
the same four numbers for every agent (%)



– random playouts miss long-term strategy

Convolutional network

the same four numbers for every agent (%)



– one bad move and the game is gone

agent 8 of 8 · search + learning

Network-guided MCTS

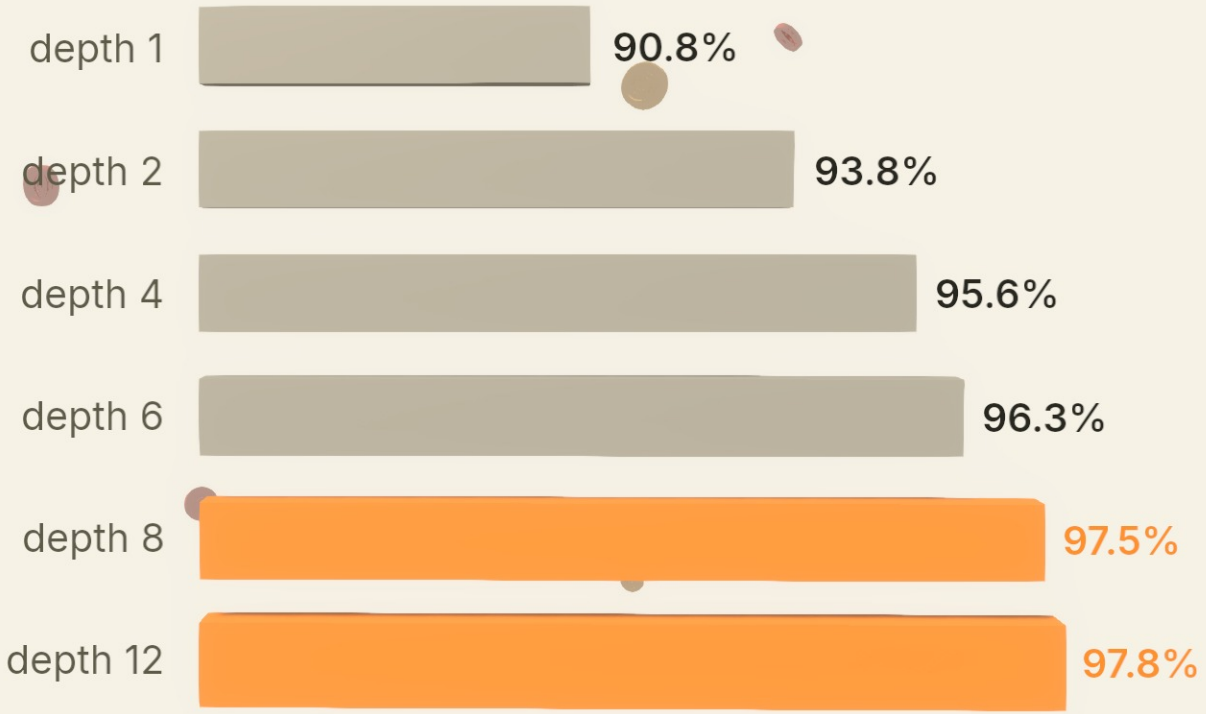
the same four numbers for every agent (%)



– each network call is slow on a CPU

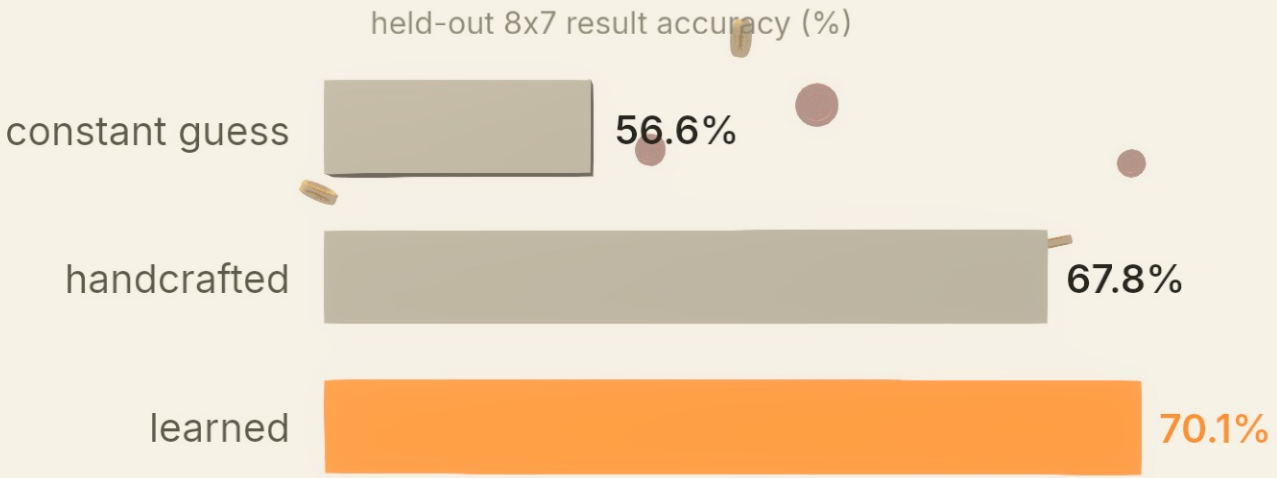
Alpha-beta: how deep?

8x7 move accuracy (%) by search depth



deeper still wins 2 of 3 games on 9x8

Learning the evaluation



on 9x8: Elo 722 learned vs 690 handcrafted

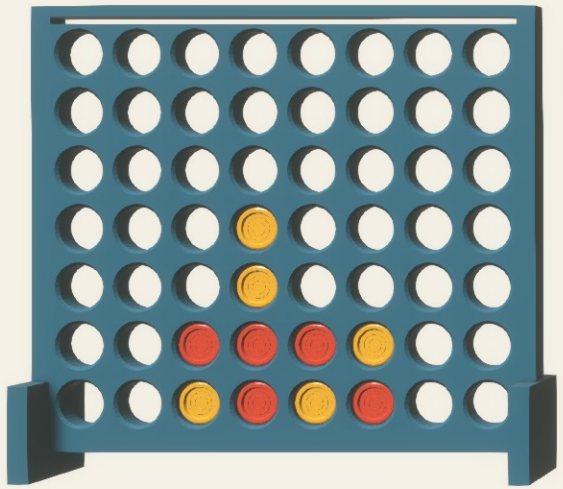
search

MCTS loves time

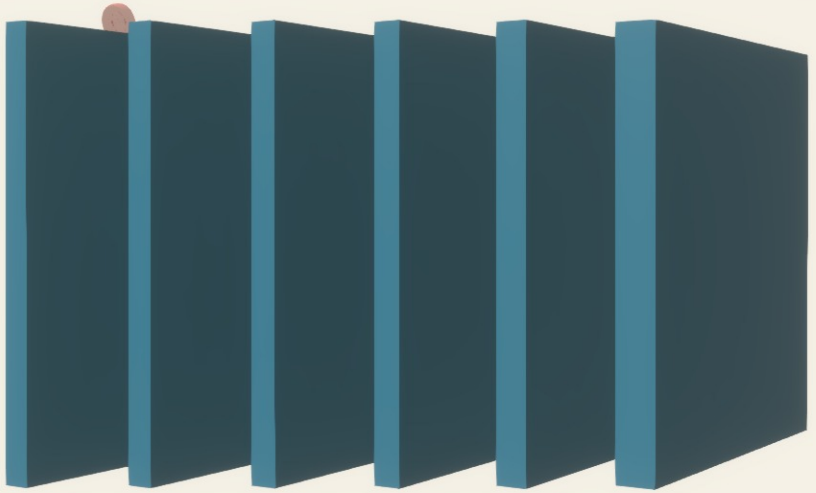
9x8 tournament Elo by time per move



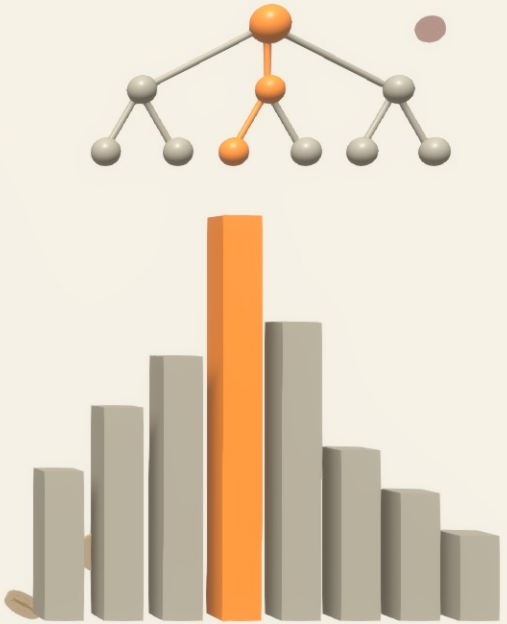
A network for any board



a board goes in



6 layers × 64 channels



a score for each column

alone: Elo 37 · guiding MCTS: Elo 647

rules

Victor Allis' strategy

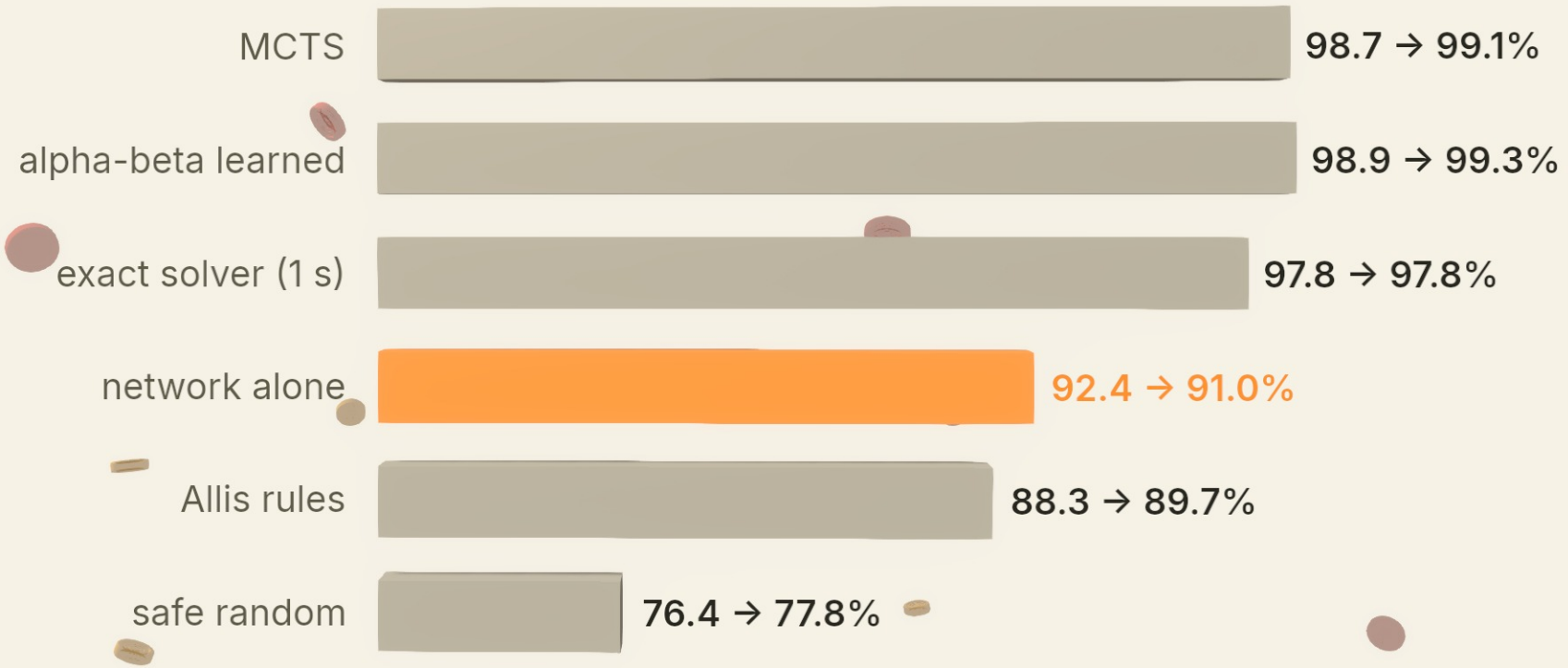
score vs safe random (%)



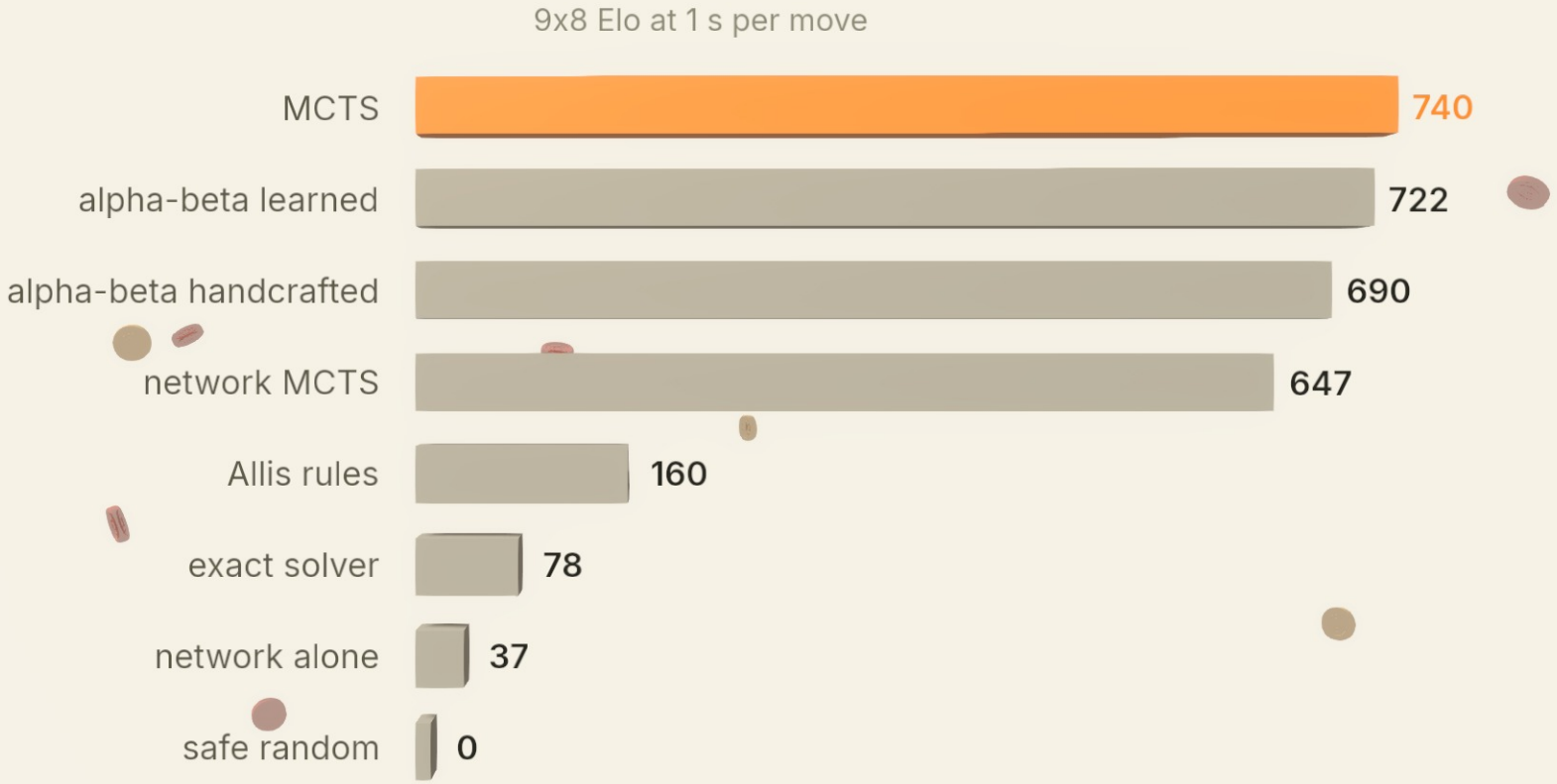
transfer

Does it carry over?

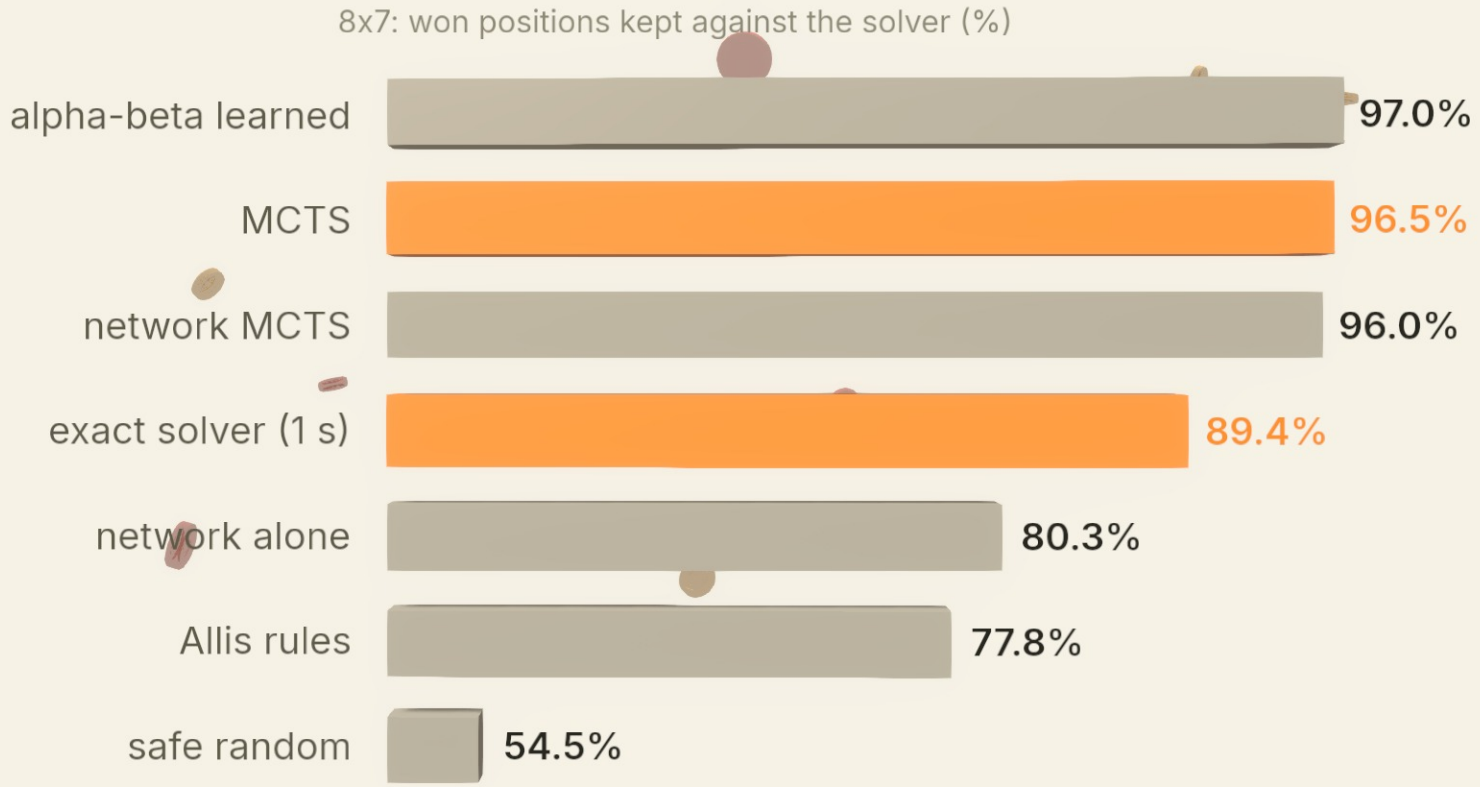
move accuracy against the solver (%): 8x7, then 9x8 late game



The 9x8 tournament

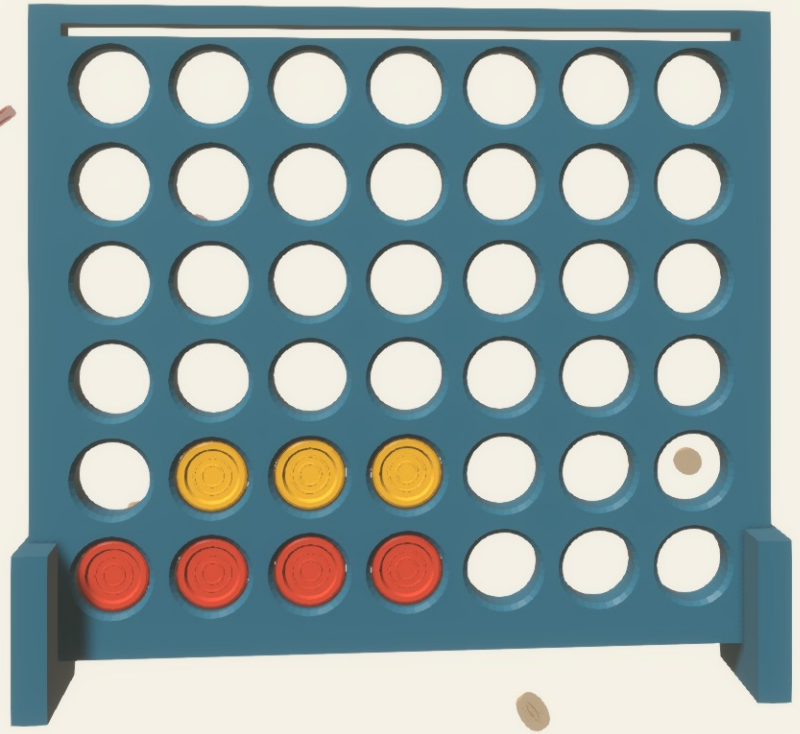


Why it came out this way



search + knowledge wins

What we learned



knowledge guides search

demo

Now watch them play

